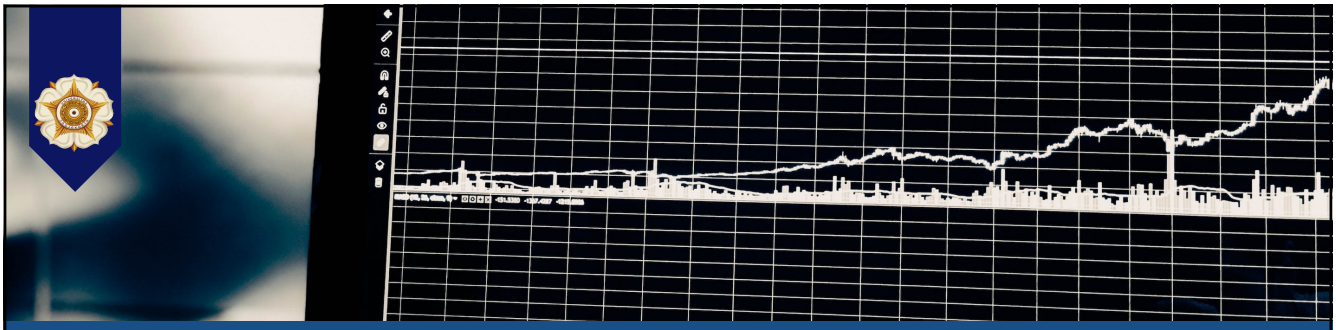


Random Sampling and Estimation (Part 1)

Sunu Wibirama
sunu@ugm.ac.id

Department of Electrical and Information Engineering
Faculty of Engineering
Universitas Gadjah Mada
INDONESIA

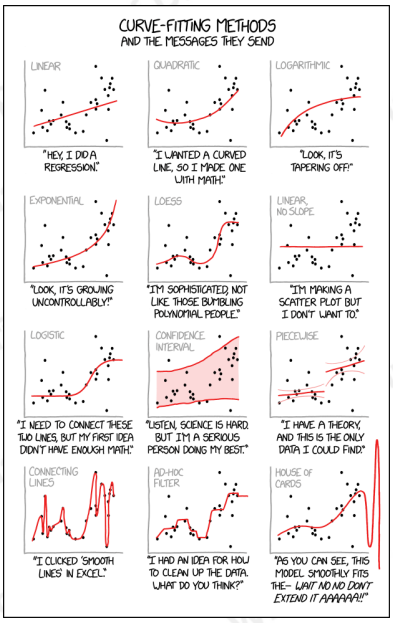
Version: August 2024

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

1

Overview of today's lecture

- Random sampling
- Central limit theorem
- Estimation of population: large sample size
- Estimation of population: small sample size



Copyright © 2024 Sunu Wibirama | Do not distribute without permission

2

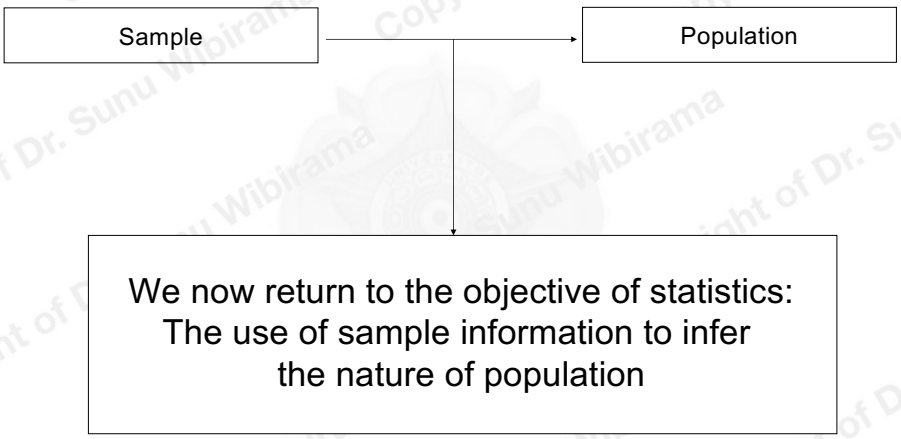
Part 1

RANDOM SAMPLING

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

3

Why the method of sampling is important ?



Copyright © 2024 Sunu Wibirama | Do not distribute without permission

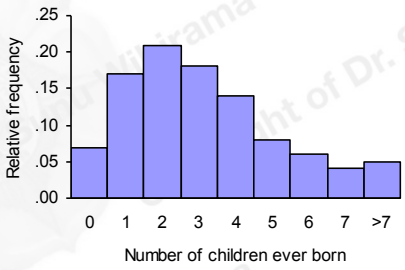
4

Data of population

Table 6.1 *Frequency distribution of number of children ever born for 4,171 women*

Number of Children	Frequency	Relative Frequency
0	312	0.07
1	708	0.17
2	881	0.21
3	737	0.18
4	570	0.14
5	354	0.08
6	243	0.06
7	172	0.04
>7	194	0.05
Total	4171	1.00

Figure 6.1 *Relative frequency distribution of number of children ever born for 4,171 women*



The Vietnam Demographic and Health Survey (VNDHS) was a nationwide representative sample survey conducted in May 1988 to collect data on fertility and a few indicators of child and maternal health.

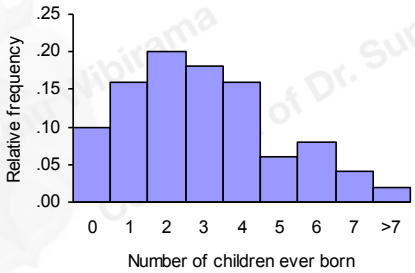
Copyright © 2024 Sunu Wibirama | Do not distribute without permission

5

Sample A: 50 data

Suppose, we randomly take 50 data from the population

Number of Children	Frequency	Relative Frequency
0	5	0.10
1	8	0.16
2	10	0.20
3	9	0.18
4	8	0.16
5	3	0.06
6	4	0.08
7	2	0.04
>7	1	0.02
Total	50	1.00



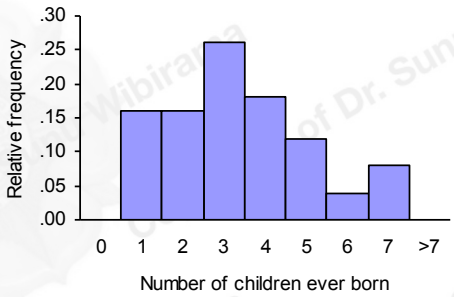
Copyright © 2024 Sunu Wibirama | Do not distribute without permission

6

Sample B: 50 data

Again for the second time, we randomly take 50 data from the population

Number of Children	Frequency	Relative Frequency
0	0	0.00
1	8	0.16
2	8	0.16
3	13	0.26
4	9	0.18
5	6	0.12
6	2	0.04
7	4	0.08
Total	50	1.00



Copyright © 2024 Sunu Wibirama | Do not distribute without permission

7

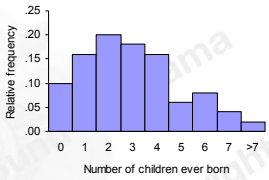
Two different samples

SAMPLE A

Table 6.2 *Frequency distribution of number of children ever born for each of two samples of 50 women selected from 4,171 women*

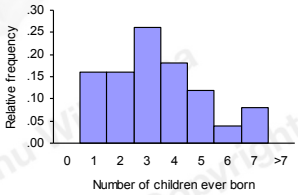
Number of Children	Frequency	Relative Frequency
0	5	0.10
1	8	0.16
2	10	0.20
3	9	0.18
4	8	0.16
5	3	0.06
6	4	0.08
7	2	0.04
>7	1	0.02
Total	50	1.00

Figure 6.2 *Frequency distribution of number of children ever born for each of two samples of 50 women selected from 4,171 women*



SAMPLE B

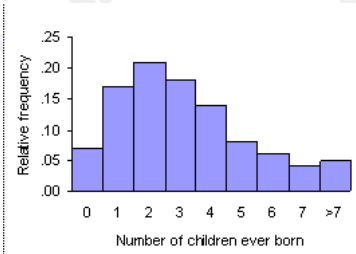
Number of Children	Frequency	Relative Frequency
0	0	0.00
1	8	0.16
2	8	0.16
3	13	0.26
4	9	0.18
5	6	0.12
6	2	0.04
7	4	0.08
Total	50	1.00



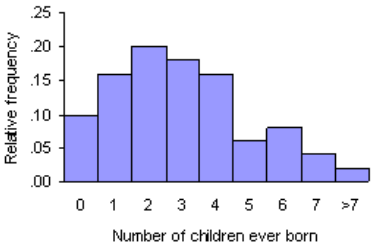
Copyright © 2024 Sunu Wibirama | Do not distribute without permission

8

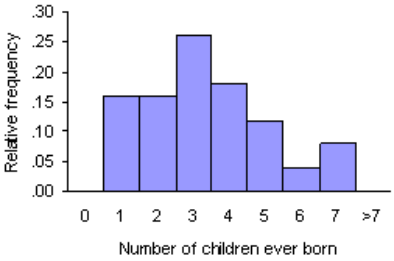
Compare both of two samples with this population graphic Which one is likely?



Population



Sample A



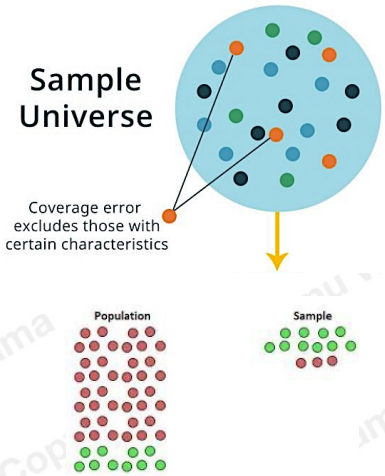
Sample B

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

9

Comparing two samples: A and B

- How is it possible: two samples of same population provides contradictory information?
- The key issue: the method by which the samples are obtained
- Great care must be taken to give an unbiased picture of the population.
- One way to cope with this problem is to use random sampling.



Copyright © 2024 Sunu Wibirama | Do not distribute without permission

10

Random sample of size n

Definition 1:

- A random sample of *n* experimental units is one selected in such a way that every different sample of size *n* has an **equal probability of selection.**

Copyright © 2024 Sunu Wibirama | Do not distribute without permission
@sunu_wibirama sunu@ugm.ac.id

11

Sampling Distribution

Definition 2:

- A numerical descriptive measure of a population is called a **parameter**.

Definition 3:

- A quantity computed from the observations in a random sample is called a **statistic**.

Definition 4:

- A sampling distribution of a sample statistic (based on *n* observations) is the relative frequency distribution of the values of the statistic theoretically generated by taking **repeated random samples of size *n*** and computing the value of the statistic for each sample.

Copyright © 2024 Sunu Wibirama | Do not distribute without permission
@sunu_wibirama sunu@ugm.ac.id

12

Parameter and statistic

- You may have observed that the value of a population parameter (for example, the population mean) is a constant (although it is usually unknown to us); its value does not vary from sample to sample.
- However, the value of a sample statistic (for example, the sample mean) is highly dependent on the particular sample that is selected.

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

13

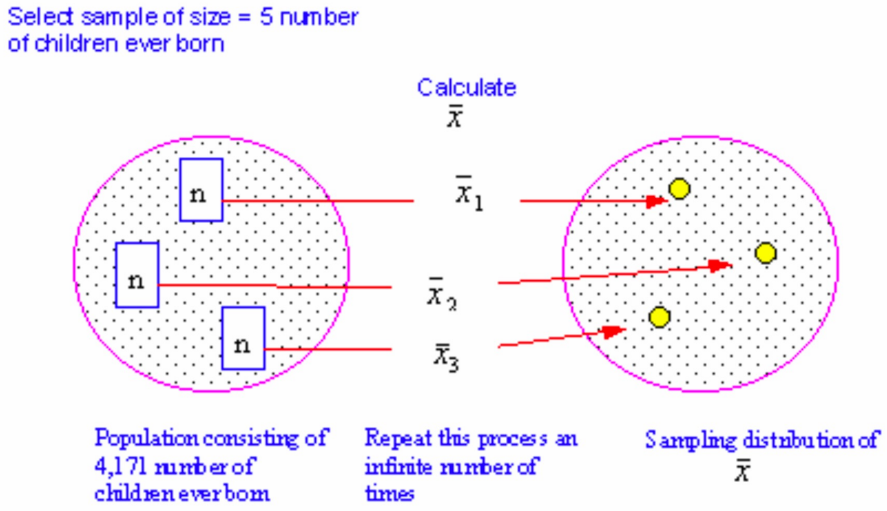
Generating sampling distribution

- How could we generate the sampling distribution of the mean of a random sample of n = 5 observations from population of 4,171 numbers of children ever born in previous slide?
  - Select a random sample of **five measurements** from the population of 4,171 observations on number of children ever born.
  - Compute and record the value of for this sample.
  - Then return these five measurements to the population and repeat the procedure.

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

14

Sampling distribution



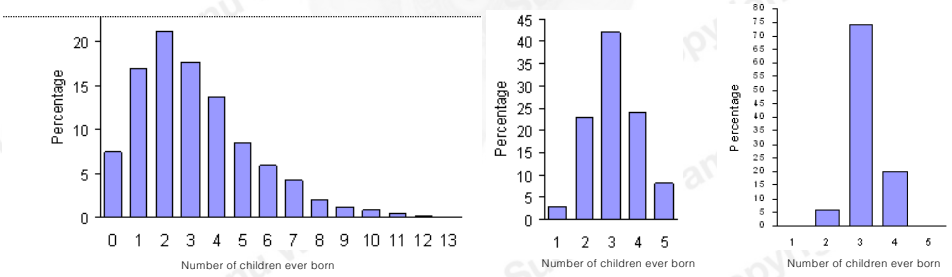
Copyright © 2024 Sunu Wibirama | Do not distribute without permission

15

Data from observation

Table 6.5 *Comparison of the population distribution and the approximate sampling distributions of $\bar{x}$, based on 100 samples of size n = 5 and n = 25*

	Mean	Standard Deviation
Population of 4,171 numbers of children ever born (Fig. 6.4)	$\mu = 3.15$	$\sigma = 2.229$
100 values of $\bar{x}$ based on samples of size $n = 5$ (Fig. 6.5)	3.11	.920
100 values of $\bar{x}$ based on samples of size $n = 25$ (Fig. 6.6)	3.14	.492



Copyright © 2024 Sunu Wibirama | Do not distribute without permission

16

Conclusion of sampling distribution

- From data above:
  - As the sample size increases, there is less variation in the sampling distribution
  - The values tend to cluster more closely about the population mean as *n* gets larger.

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

17

Part 2

CENTRAL LIMIT THEOREM

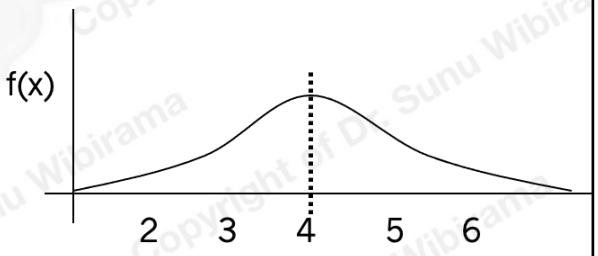
Copyright © 2024 Sunu Wibirama | Do not distribute without permission

18

Review: normal distribution

- The most important continuous probability distribution
- The graph called the “normal curve” (bell-shaped).
- Total area under the curve = 1
- Derived by DeMoivre and Gauss. The distribution is also called the **“Gaussian” distribution.** It describes many phenomena in nature, industry and research.

$$f(x) = n(x;\mu,\sigma) = \frac{e^{-0.5\left[\frac{(x-\mu)}{\sigma}\right]^2}}{\sigma\sqrt{2\pi}}, \qquad -\infty < x < \infty$$



Copyright © 2024 Sunu Wibirama | Do not distribute without permission

19

The central limit theorem

- If the size is sufficiently large, the mean of a random sample from a population has a sampling distribution that is approximately normal*, regardless of the shape of the relative frequency distribution of the target population.
- As the sample size increases, the better will be the normal approximation to the sampling distribution

*remember: normal distribution

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

20

The central limit theorem

Properties of Sampling Distribution of $\bar{x}$

If $\bar{x}$ is the mean of a random sample of size n from a population with mean μ and standard deviation σ, then:

1. The sampling distribution of $\bar{x}$ has a mean equal (almost equal) to the mean of the population from which the sample was selected. That is, if we let $\mu_{\bar{x}}$ denote the mean of the sampling distribution of $\bar{x}$, then

$$\mu_{\bar{x}} = \mu$$

2. The sampling distribution of $\bar{x}$ has a standard deviation equal to the standard deviation of the population from which the sample was selected, divided by the square root of the sample size. That is, if we let $\sigma_{\bar{x}}$ denote the standard deviation of the sampling distribution of $\bar{x}$, then

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

21

The central limit theorem

(from Walpole's book p. 234)

Theorem 8.2: **Central Limit Theorem:** If $\bar{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2, then the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

as $n \to \infty$, is the standard normal distribution $n(z; 0, 1)$.

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

22

The central limit theorem

(from Walpole's book p. 234)

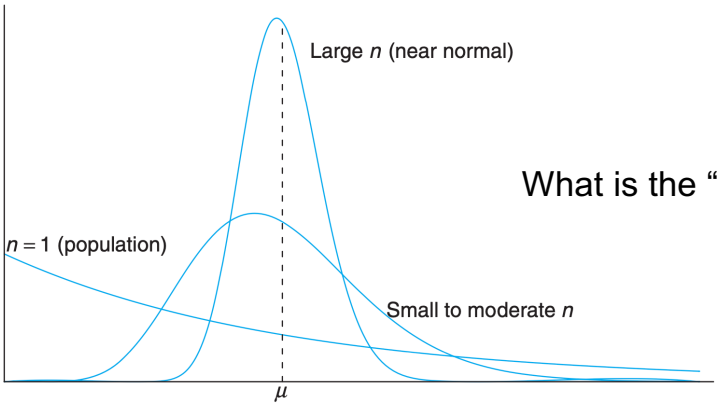


Figure 8.1: Illustration of the Central Limit Theorem (distribution of $\bar{X}$ for $n = 1$, moderate n, and large n).

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

23

The central limit theorem

(from Walpole's book p. 234)

The normal approximation for $\bar{X}$ will generally be good if $n \geq 30$, provided the population distribution is not terribly skewed. If $n < 30$, the approximation is good only if the population is not too different from a normal distribution and, as stated above, if the population is known to be normal, the sampling distribution of $\bar{X}$ will follow a normal distribution exactly, no matter how small the size of the samples.

The sample size $n = 30$ is a guideline to use for the Central Limit Theorem. However, as the statement of the theorem implies, the presumption of normality on the distribution of $\bar{X}$ becomes more accurate as n grows larger. In fact, Figure 8.1 illustrates how the theorem works. It shows how the distribution of $\bar{X}$ becomes closer to normal as n grows larger, beginning with the clearly nonsymmetric distribution of an individual observation ($n = 1$). It also illustrates that the mean of $\bar{X}$ remains μ for any sample size and the variance of $\bar{X}$ gets smaller as n increases.

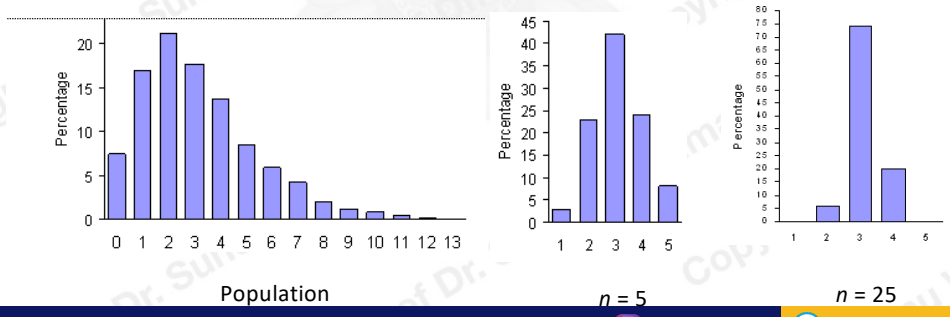
Copyright © 2024 Sunu Wibirama | Do not distribute without permission

24

Example: previous data

Table 6.5 *Comparison of the population distribution and the approximate sampling distributions of $\bar{x}$, based on 100 samples of size n = 5 and n = 25*

	Mean	Standard Deviation
Population of 4,171 numbers of children ever born (Fig. 6.4)	$\mu = 3.15$	$\sigma = 2.229$
100 values of $\bar{x}$ based on samples of size $n = 5$ (Fig. 6.5)	3.11	.920
100 values of $\bar{x}$ based on samples of size $n = 25$ (Fig. 6.6)	3.14	.492



Copyright © 2024 Sunu Wibirama | Do not distribute without permission

25

Example: previous data

- Show that the empirical evidence obtained in previous observation supports the Central Limit Theorem and two properties of the sampling distribution of $\bar{x}$.
- For this target population, we know that the values of the parameters are:
- *Population mean:* *= 3.15*
  Population standard deviation *= 2.229*

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

26

Answer

- For *n*=5, the Standard Deviation:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{2.229}{\sqrt{5}} = .997$$

Simulation result: .920

- For *n*=25, the Standard Deviation:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{2.229}{\sqrt{25}} = .446$$

Simulation result: .492

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

27

Example (Walpole's book)

Example 8.5: Traveling between two campuses of a university in a city via shuttle bus takes, on average, 28 minutes with a standard deviation of 5 minutes. In a given week, a bus transported passengers 40 times. What is the probability that the average transport time was more than 30 minutes? Assume the mean time is measured to the nearest minute.

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

28

Example (Walpole's book)

Example 8.5: Traveling between two campuses of a university in a city via shuttle bus takes, on average, 28 minutes with a standard deviation of 5 minutes. In a given week, a bus transported passengers 40 times. What is the probability that the average transport time was more than 30 minutes? Assume the mean time is measured to the nearest minute.

Remember:

Central Limit Theorem: If $\bar{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2, then the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

as $n \to \infty$, is the standard normal distribution $n(z; 0, 1)$.

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

29

Review: Finding *x* from *z*

1. **Draw a picture of the distribution.**
2. **Translate the problem into one of the following:**

   $$P(X < a), \quad P(X > b), \quad \text{or} \quad P(a < X < b).$$

   Shade in the area on your picture.
3. **Transform a (and/or b) into a z-value, using the Z-formula:**

   $$Z = \frac{X - \mu}{\sigma}.$$

4. **Look up the transformed z-value on the Z-table (in the Walpole's book, see the Table A.3) and find its probability.**

5a. **If you have a less-than problem, you are done.**

5b. **If you have a greater-than problem, take one minus the result from Step 4.**

5c. **If you have a between-values problem, do Steps 1–4 for b (the larger of the two values) and then for a (the smaller of the two values), and subtract the results.**

Table A.3 Normal Probability Table 755



Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

30

Example (Walpole's book)

Example 8.5: Traveling between two campuses of a university in a city via shuttle bus takes, on average, 28 minutes with a standard deviation of 5 minutes. In a given week, a bus transported passengers 40 times. What is the probability that the average transport time was more than 30 minutes? Assume the mean time is measured to the nearest minute.

***Solution*:** In this case, $\mu = 28$ and $\sigma = 3$. We need to calculate the probability $P(\bar{X} > 30)$ with $n = 40$. Since the time is measured on a continuous scale to the nearest minute, an $\bar{x}$ greater than 30 is equivalent to $\bar{x} \geq 30.5$. Hence,

$$P(\bar{X} > 30) = P\left(\frac{\bar{X} - 28}{5/\sqrt{40}} \geq \frac{30.5 - 28}{5/\sqrt{40}}\right) = P(Z \geq 3.16) = 0.0008.$$

There is only a slight chance that the average time of one bus trip will exceed 30 minutes. An illustrative graph is shown in Figure 8.4.

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

31

Example (Walpole's book)

***Solution*:** In this case, $\mu = 28$ and $\sigma = 3$. We need to calculate the probability $P(\bar{X} > 30)$ with $n = 40$. Since the time is measured on a continuous scale to the nearest minute, an $\bar{x}$ greater than 30 is equivalent to $\bar{x} \geq 30.5$. Hence,

$$P(\bar{X} > 30) = P\left(\frac{\bar{X} - 28}{5/\sqrt{40}} \geq \frac{30.5 - 28}{5/\sqrt{40}}\right) = P(Z \geq 3.16) = 0.0008.$$

There is only a slight chance that the average time of one bus trip will exceed 30 minutes. An illustrative graph is shown in Figure 8.4.

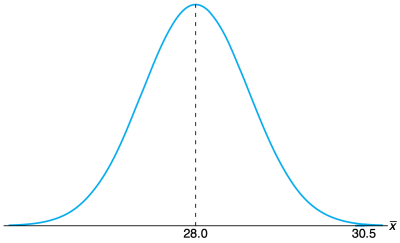


Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

$P(Z \geq 3.16) = 1 - P(Z \leq 3.16)$

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

32

Take home messages

- Today's key messages:
  - How to do sampling is important to provide fairness in your data.
  - Random sampling is a solution to avoid bias when estimating parameters of population.
  - Size of sample matters. If the size is sufficiently large, the mean of a random sample from a population has a sampling distribution that is approximately normal → explained in detail next week
  - Z-distribution is very useful to estimate probability of a particular case when we perform sampling from population.

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

33

End of File

Copyright © 2024 Sunu Wibirama | Do not distribute without permission

34